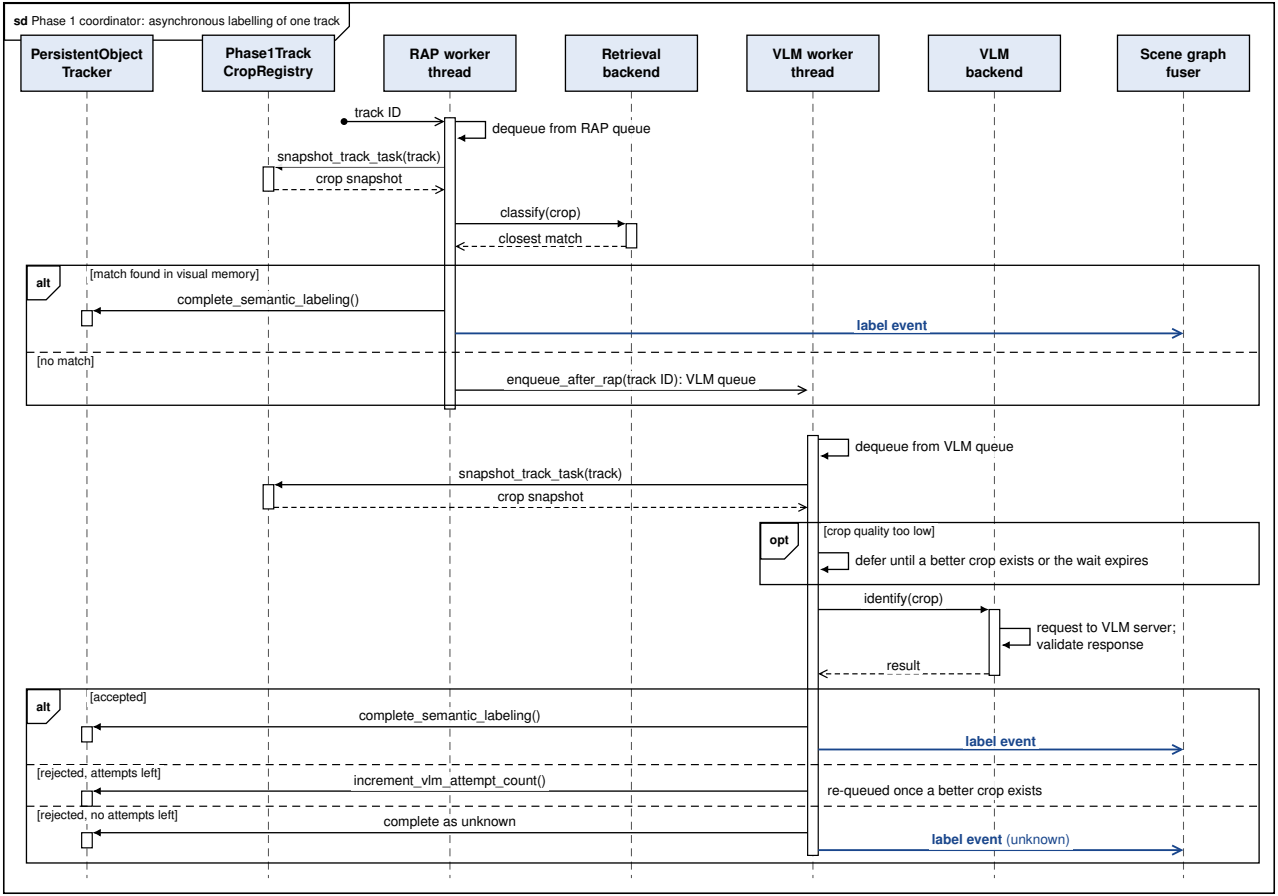
sd Phase 1 coordinator: asynchronous labelling of one track
PersistentObject Tracker
Phase1Track CropRegistry
RAP worker thread
Retrieval backend
VLM worker thread
VLM backend
Scene graph fuser
track ID
dequeue from RAP queue
snapshot_track_task(track)
crop snapshot
classify(crop)
closest match
alt
[match found in visual memory]
complete_semantic_labeling()
label event
[no match]
enqueue_after_rap(track ID): VLM queue
dequeue from VLM queue
snapshot_track_task(track)
crop snapshot
opt
[crop quality too low]
defer until a better crop exists or the wait expires
identify(crop)
request to VLM server; validate response
result
alt
[accepted]
complete_semantic_labeling()
label event
[rejected, attempts left]
increment_vlm_attempt_count()
re-queued once a better crop exists
[rejected, no attempts left]
complete as unknown
label event (unknown)